

ASTRO RAIDER

Game Design Document — v1.0 Plan

1. Game Overview

Astro Raider is a 2D Pixel Art Roguelike Endless Shooter planned for Steam. The player defends a Generator against enemy waves while mining resources, building towers, and collecting perks.

Genre	2D Pixel Art Roguelike Endless Shooter
Platform	Steam (PC)
Engine	Godot 4 / GDScript
Developer	Samy (solo)
Publisher	Samy (own name)
Price	20 EUR
Run Duration	~20 minutes
v1.0 Multiplayer	Solo only
Post-v1.0	Steam Online Multiplayer
Target Timeline	~3 months focused development

2. Lore & Story

The player is an astronaut who dies and is revived by a sarcastic Demon. The Demon demands 1 million in debt repayment as a running joke — the debt can never truly be paid. The Demon insults and guides the player throughout each run.

The Generator

The Generator is the central structure. Lore-wise it emits a field keeping monsters at bay. When destroyed, all restraints lift and the Endgame Phase begins. The Generator cannot be repaired.

Endgame Phase (Generator Destroyed)

- No new perks available (existing perks still active)
- Healing and all other gameplay still possible
- Screen gradually turns red
- Music changes to intense final theme
- Enemies spawn from all directions simultaneously
- Enemies become significantly stronger and faster
- Player fights until death — no recovery possible

3. Core Gameplay Loop

- Spawn on randomly selected pre-generated world (1 of 20)
- Defend Generator against enemy waves
- Mine resources (Gold, Iron, Copper) and collect Crystals
- Visit Generator to unlock Perks or upgrade Buildings
- Survive until Boss Wave (final wave) to complete run
- Die or complete run -> restart with Meta-Progression unlocks

Wave System

Event	Detail
Between waves	Short pause for movement/mining
Every 5 waves	Longer pause + enemies become faster/stronger
Final wave	Boss fight (Ketra or Snake)
Generator destroyed	Endgame Phase triggers immediately
Score	Waves survived

4. World & Map

Map Size	400x400 tiles (6400x6400 pixels)
Tile Size	16x16 pixels
Worlds	20 pre-generated, shipped with game
Selection	Random per run
Save Format	Binary .dat via TileMapData Resource (fast loading)
Planets v1.0	3-5 planets with unique configs

Planet Config System

Setting	Effect
Gravity Strength	Changes how the player moves and feels on each planet
TileDrops	Different resources, types and rarities per planet
Room Settings	Different world generation shape and room sizes
Noise Settings	Different organic terrain forms
Background Parallax	3-4 parallax layers for depth and immersion
Enemy Set	Planet-specific enemies and skins
Tileset / Assets	Visual theme of the planet

5. Tower Defense Elements

All buildings managed and upgraded at the Generator using collected resources (Gold, Iron, Copper).

Building	Function	Status
Generator	Core structure. Perk Selector + Upgrade hub. Endgame Phase if destroyed.	Implemented
Turret	Shoots at nearby enemies automatically.	Implemented
Platform	Provides additional terrain and positioning.	Implemented
Repair Drone	Repairs nearby structures.	Implemented

6. Perk System

Category	Max Active	Levels	Rarity Weight	Button	Notes
Passive	Unlimited	6	10	—	Most common
Activation	3 slots	6	5	Q / E / C	Locked to slots
Ult	1 slot	3	2	X	Rarest, most powerful

Ult Balancing

- Ult Level 1 = equivalent to Passive Level 6
- Ult Level 3 = equivalent to Passive Level 18

Slot Assignment UI

- After selecting Activation Perk, empty slots light up and pulse in perk color
- Player presses Q/E/C key OR clicks the slot in HUD
- Perk animates into slot with scale tween
- No slot replacement allowed

Selector Rules

- Triggered at Generator by button (requires Crystals)
- First Ult selection: show only 3 Ults
- First Activation selection per empty slot: show only Activation Perks
- After all slots filled: all perk types mixed for leveling
- All 3 Activation slots full -> no new Activation Perks offered
- 1 Ult selected -> no more Ults offered
- Exclusion System: perks can block other perks (set in Inspector)
- Rarity Weight: higher = appears more often

7. Enemies

Enemy	Behavior	Skins	Status
Bat	Dash toward player	3 variants	Implemented
Flying Eye	Dash + Ranged attack	3 variants	Implemented
Fat Slime	Tank, floats (space lore)	3 variants	Implemented

! Each planet will have its own enemy set and skins. Stun system implemented.

Planned: Enemy Bases

Destroyable structures that spawn enemies. Destroying them yields bonus resources. Adds offensive objective alongside Generator defense.

8. Bosses

Boss is the final wave of each run. Boss Bar implemented and animated. Boss triggers win condition.

Boss	Description	Mechanic	Status
Ketra	Large spider with blade arms and single eye	Multi-phase fight, eye is weak point	Designed, not implemented
Snake	Segmented serpent body with shark-like head	Segments are individually destroyable	Designed, not implemented

9. Meta-Progression & Shop

- Between runs, players permanently unlock new Perks
- Unlocked Perks appear in Selector in future runs
- Shop: buy Perks using run currency (currency type TBD)
- No mid-run saving for v1.0 — always restart fresh
- Steam Achievements planned

10. Tutorial & UI

Tutorial

- Optional and skippable interactive tutorial
- Demon guides player with sarcastic insults
- Dialogs run WITHOUT pausing the game
- Visual UI hints shown independently from dialog
- Implemented: Movement tutorial, Shooting tutorial
- Planned: Full overhaul in Tutorial Crash Course

UI Overhaul (Planned)

- Lifebar redesign
- Full HUD optimization
- Perk slot display (Q/E/C/X) always visible
- Building upgrade UI at Generator
- Wave counter
- Resource display (Gold, Iron, Copper, Crystals)

11. Technical Architecture

Environment System

- Enviroment.gd — coordinator, signals, map generation trigger
- EnvironmentGenerator.gd — chunk/room/noise generation
- EnvironmentTiles.gd — tile destruction, drops, refresh
- EnvironmentPersistence.gd — binary save/load via TileMapData Resource
- EnvironmentReveal.gd — wall item notification system

Player System

- Player.gd — CharacterBody2D coordinator
- Movement.gd — 4-direction gravity, get_input_direction() for keyboard/controller
- Mining.gd — bohrer state, tile destruction, static hit damage
- Build.gd — building mode toggle

Perk System

- PerkBuild.gd — base class, extend and override activate_perk()
- PerkManager.gd — selection, cooldown, exclusion, ult/slot tracking
- PerkSelector.gd — hold-to-select UI, dynamic button count
- Perk.gd — Resource: value[6], cost[6], active_type, excluded_perks, rarity_weight

Projectile System

- PlayerProjectile.gd — Area2D, HP, AimBot, damage/crit/stun passthrough
- EnemyDetectorArea.gd — enemy tracking, closest target, lock-on after hit
- Trail.gd — MeshInstance2D, ImmediateMesh, top_level=true, custom UVs
- trail_shader.gdshader — 3-color pulse shader, core/glow effect, blend_add
- TODO: Object pooling for projectiles
- TODO: Trail shader final refinement

Global Systems

- GSignals — global signal bus for decoupled communication
- GlobalGame — shared player/game state
- PlayerDataBuilds — player build persistence
- DropData — item drop registry
- PerkData — perk resource registry

12. Priority List

RED [!!] = Must have for v1.0 | ORANGE [!] = Important | BLUE [~] = Post-v1.0

CRITICAL — Must ship with v1.0

- [!!] Boss implementation (Ketra or Snake as final wave)
- [!!] Perk Slot System Q/E/C/X with slot assignment UI
- [!!] Perk Selector trigger at Generator (button)
- [!!] Ult selection screen (show 3 Ults on first selection)
- [!!] Endgame Phase when Generator destroyed
- [!!] 2+ additional planets with unique enemies and gravity
- [!!] Tutorial overhaul (non-pausing Demon dialogs)
- [!!] UI/HUD overhaul (lifebar, perk slots, resources)
- [!!] Difficulty scaling fix (every 5 waves)
- [!!] Basic Meta-Progression (unlock perks between runs)
- [!!] Steam page + Achievements

IMPORTANT — Should ship with v1.0

- [!] Building upgrade UI at Generator
- [!] Rarity weight system for perk selector
- [!] Enemy bases (destroyable, bonus resources)
- [!] Shop system for unlocking perks
- [!] Sound/music optimization
- [!] Parallax backgrounds per planet
- [!] PlanetConfig Resource system
- [!] Object pooling for projectiles

NICE TO HAVE — Post-v1.0

- [~] Online Multiplayer via Steam
- [~] Weapon system (changes perk behavior per weapon)
- [~] Trail shader final refinement
- [~] Additional enemy types beyond 3 per planet
- [~] Steam Playtest phase before launch
- [~] DaVinci Resolve trailer (record clips during development!)

13. Development Timeline

Based on 18h/week (2h weekdays + 4h weekends). ~70h total with AI assistance.

Week	Focus	Tasks	Hours	Milestone
1	Perk Slots	Q/E/C/X system, slot UI, Perk Selector at Generator, Ult screen	10h	—
2	Planets	PlanetConfig, Planet 2 (assets, enemies, gravity, parallax)	14h	—
3	Planet 3 + Boss	Planet 3, begin Boss implementation	14h	—
4	Boss + Endgame	Finish Boss, Endgame Phase, difficulty scaling fix	14h	ALPHA
5	Meta + Shop	Meta-Progression, Shop, Rarity weights	12h	—
6	UI + Tutorial	HUD overhaul, lifebar, perk slots, Tutorial overhaul	14h	—

Week	Focus	Tasks	Hours	Milestone
7	Polish	Building upgrade UI, enemy bases, object pooling, sounds	14h	BETA
8	Buffer	Bug fixing, balance, edge cases, Steam page	14h	RC
9-10	Playtest	Discord Playtest, Steam Playtest, fixes	20h	PLAYTEST
11-12	Final	Last fixes, Achievements, trailer clips, launch prep	14h	v1.0

! Buffer in weeks 8-10 is intentional. Never skip buffer weeks.

14. Key Milestones

Milestone	Description	Target
Alpha	All core systems: Perks, Boss, Endgame Phase, 3 Planets	Week 4
Beta	All content: Meta-Prog, Shop, UI, Tutorial, 5 Planets	Week 7
Release Candidate	Bug-free build ready for Playtest	Week 8
Playtest	Discord + Steam Playtest open	Week 10
v1.0 Release	Steam launch after playtest feedback integrated	Week 12

15. Planned Crash Courses

Session	Scope	Target Week
Perk Crash Course	Review 14 existing perks, assign categories and rarity weights, plan remaining ~11	Week 1-2
KI/Enemy Crash Course	Improve AI behaviors, add new enemy types per planet, fix difficulty scaling code	Week 2-3
Tutorial Crash Course	Review current tutorial, expand Demon dialogs, non-pausing dialog system	Week 6
UI Crash Course	Redesign HUD, lifebar, perk slot display (Q/E/C/X), building upgrade UI	Week 6
Run Structure Crash Course	Define exact wave count, difficulty curve, boss trigger logic	Week 4

16. v1.0 Checklist

Implemented

- [x] World generation (400x400, 20 worlds, binary save/load)
- [x] 3 enemy types with multiple skins
- [x] 14 perks (6 levels each)
- [x] Tower Defense (Generator, Turret, Platform, Repair Drone)
- [x] Wave system with pauses (every 5 waves: harder + longer pause)
- [x] Crystal/perk unlock system
- [x] Mining and resource system (Gold, Iron, Copper)

- [x] Bomb system
- [x] 4-direction gravity movement
- [x] AimBot perk for projectiles (lock-on after hit)
- [x] Boss Bar (animated)
- [x] 2 bosses designed (Ketra, Snake)
- [x] Multiple planet backgrounds (random selection)
- [x] Minimap + Radar
- [x] Demon tutorial dialogs
- [x] Local multiplayer camera base
- [x] Sounds and music (needs optimization)
- [x] Trail system (MeshInstance2D + shader)
- [x] Environment split into 5 components
- [x] Perk exclusion + cooldown system

Remaining for v1.0

- [] Boss implementation (Ketra or Snake)
- [] Perk Slot System Q/E/C/X
- [] Slot assignment UI (slots light up, press key or click)
- [] Perk Selector trigger at Generator
- [] Ult selection screen
- [] Endgame Phase (generator destroyed)
- [] 2+ additional planets
- [] Planet Config Resource system
- [] Parallax backgrounds
- [] Gravity per planet
- [] Tutorial overhaul
- [] UI/HUD overhaul (lifebar, perk slots)
- [] Difficulty scaling fix
- [] Meta-Progression system
- [] Shop system
- [] Building upgrade UI
- [] Rarity weight system
- [] Enemy bases
- [] Object pooling for projectiles
- [] Sound/music optimization
- [] Steam Achievements
- [] Steam page setup

17. Revenue Estimate

Scenario	Sales	Gross (20 EUR)	Net after Steam+Tax (~50%)
Pessimistic	200-500	4.000-10.000 EUR	2.000-5.000 EUR
Realistic	500-2.000	10.000-40.000 EUR	5.000-20.000 EUR
Optimistic	2.000-5.000	40.000-100.000 EUR	20.000-50.000 EUR

! Steam Direct Fee: 100 EUR (refunded after 1000 USD revenue).

- Record gameplay clips during development — use for trailer later

- Create Coming Soon page early to collect wishlists
- Run Discord Playtest first, then Steam Playtest
- Tags: Roguelike, Bullet Hell, Tower Defense, Pixel Art, Singleplayer
- Target Youtubers/streamers who cover Roguelikes

Astro Raider GDD v4 — Dark Edition